

CLAUSES AND AGGREGATION IN MYSQL :

In MySQL (as well as in many other relational database management systems), clauses and aggregations are key concepts used in constructing queries to retrieve and manipulate data.

Clauses:

In SQL, a clause is a component or a part of a SQL statement that defines the conditions of the query. Some commonly used clauses include:

SELECT Clause: Specifies the columns you want to retrieve in the result set.

Sql

SELECT column1, column2 FROM table_name;

FROM Clause: Indicates the table or tables from which to retrieve data.

sql

SELECT column1, column2 FROM employees;

WHERE Clause: Filters the rows returned based on a specified condition.

sql

```
SELECT column1, column2 FROM employees WHERE salary > 50000;
```

ORDER BY Clause: Sorts the result set based on specified columns and in a specified order.

sql

```
SELECT column1, column2 FROM employees ORDER BY salary DESC;
```

GROUP BY Clause: Groups rows that have the same values in specified columns into summary rows.

sql

```
SELECT department, AVG(salary) FROM employees GROUP BY department;
```

HAVING Clause: Filters the results of a GROUP BY clause based on specified conditions.

Sql

SELECT department, AVG(salary) FROM employees GROUP BY department HAVING AVG(salary) > 60000;

AGGREGATIONS:

Aggregations are used to perform calculations on a set of values and return a single value. Common aggregation functions include:

COUNT(): Counts the number of rows in a result set.

sql

SELECT COUNT() FROM employees;*

SUM(): Calculates the sum of values in a numeric column.

sql

SELECT SUM(salary) FROM employees;

AVG(): Calculates the average of values in a numeric column.

sql

SELECT AVG(salary) FROM employees;

MIN(): Retrieves the minimum value in a column.

sql

```
SELECT MIN(salary) FROM employees;
```

MAX(): Retrieves the maximum value in a column.

sql

```
SELECT MAX(salary) FROM employees;
```

These clauses and aggregation functions are fundamental for constructing powerful and flexible SQL queries in MySQL. They allow you to filter, sort, group, and aggregate data according to your specific requirements.